

# VPC Peering.

## 1. What VPC Peering Actually Is

VPC Peering is a network connection between two Virtual Private Clouds (VPCs) that allows resources in both VPCs to communicate privately using private IP addresses.

No internet gateway.

No NAT traversal.

Traffic stays inside AWS network infrastructure.

In simple terms:

Two isolated networks become directly connected.

Example:

- VPC-A: 10.0.0.0/16
- VPC-B: 192.168.0.0/16

After peering, an EC2 instance in VPC-A can directly reach an EC2 in VPC-B.

## 2. Why VPC Peering Exists

AWS isolates VPC networks by default. Without peering, VPCs cannot communicate.

Peering is used when you want:

- Communication between multiple environments
- Communication between different AWS accounts
- Communication between different regions

Common cases:

- Dev VPC ↔ Production VPC
- Application VPC ↔ Database VPC
- Shared services VPC ↔ Multiple project VPCs

### 3. Types of VPC Peering

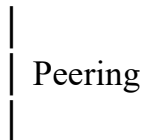
#### Intra-Region VPC Peering

Both VPCs are in the same AWS region.

Example

Mumbai region:

VPC-A (10.0.0.0/16)



VPC-B (172.16.0.0/16)

Characteristics:

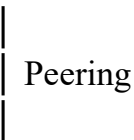
- Low latency
- No cross-region cost
- Simple routing

#### Inter-Region VPC Peering

VPCs are in different AWS regions.

Example:

VPC-A (Mumbai)



VPC-B (Singapore)

AWS encrypts the traffic automatically.

Used for:

- Multi-region architectures

- Disaster recovery setups

I have created one VPC in the N.virginia region.

**VPC settings**

**Resources to create** [Info](#)  
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

**Name tag - optional** [Info](#)  
Creates a tag with a key of 'Name' and a value that you specify.

myvpc

**IPv4 CIDR block** [Info](#)  
☒ IPv4 CIDR manual input  
☐ IPAM-allocated IPv4 CIDR block

**IPv4 CIDR**  
10.0.0.0/16  
CIDR block size must be between /16 and /28.

**IPv6 CIDR block** [Info](#)  
☒ No IPv6 CIDR block  
☐ IPAM-allocated IPv6 CIDR block  
☐ Amazon-provided IPv6 CIDR block  
☐ IPv6 CIDR owned by me

**Tenancy** [Info](#)  
Default

**VPC encryption control (\$)** [Info](#)  
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply [Link](#).

☒ None ☐ Monitor mode ☐ Enforce mode

The other in the Mumbai region.

**Create VPC** [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

**VPC settings**

**Resources to create** [Info](#)  
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☒ VPC only ☐ VPC and more

**Name tag - optional** [Info](#)  
Creates a tag with a key of 'Name' and a value that you specify.

myvpc1

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**Tenancy** [Info](#)  
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And I have configured all the recommended settings for the VPC to Activate.

(i.e) internet gateway , security groups , subnets.

The creation of public instance in the N.virginia region.

aws  Search [Alt+S] United States (N. Virginia) Dhanam M

EC2 > Instances > Launch an instance

Key pair name - required  
hari  [Create new key pair](#)

▼ Network settings Info

VPC - required Info  
vpc-084f2a0a522890979 (myvpc)  

Subnet Info  
subnet-07fdb28678267d71  [Create new subnet](#) 

VPC: vpc-084f2a0a522890979 Owner: 051741041276 Availability Zone: us-east-1a (use1-az1)  
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP Info  
Enable 

Firewall (security groups) Info  
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.  
☐ Create security group ☒ Select existing security group

Common security groups Info  
Select security groups  [Compare security group rules](#)

publicsg sg-023841209409f2738   
VPC: vpc-084f2a0a522890979

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Number of instances Info  
1

Software Image (AMI)  
Amazon Linux 2023 AMI 2023.10...[read more](#)  
ami-02dfbd4ff395f2a1b

Virtual server type (instance type)  
t3.micro

Firewall (security group)  
publicsg

Storage (volumes)  
1 volume(s) - 8 GiB

Cancel [Launch instance](#) [Preview code](#)

The creation of public instance in the Mumbai region.

▼ Network settings Info

VPC - required Info  
vpc-084f2a0a522890979 (myvpc)  

Subnet Info  
subnet-07fdb28678267d71  [Create new subnet](#) 

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Cancel [Launch instance](#) [Preview code](#)

Before peering we have to give one more route values to all the VPC's.

1<sup>st</sup> VPC.

aws  Search [Alt+S] United States (N. Virginia) Dhanam M

VPC > Route tables > rtb-0549d12b52882b56b > Edit routes

Edit routes

| Destination                                                                                                                                                                         | Target                                                                                                                                                                                        | Status                                                                                     | Propagated | Route Origin                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------|------------------------------------|
| 10.0.0.0/16                                                                                                                                                                         | local                                                                                                      | Active  | No         | CreateRouteTable                   |
|  0.0.0.0/0    |  local                  | -                                                                                          | No         | CreateRoute <a href="#">Remove</a> |
|  11.0.0.0/16  |  Internet Gateway       | -                                                                                          | No         | CreateRoute <a href="#">Remove</a> |
|  11.0.0.0/16  |  igw-00fc645da45d57b27  | -                                                                                          | No         | CreateRoute <a href="#">Remove</a> |
|  11.0.0.0/16  |  Peering Connection     | -                                                                                          | No         | CreateRoute <a href="#">Remove</a> |
|  11.0.0.0/16  |  pcx-01646ef6965f09fef  | -                                                                                          | No         | CreateRoute <a href="#">Remove</a> |

[Add route](#)

Cancel [Preview](#) [Save changes](#)

2<sup>nd</sup> VPC.

The screenshot shows the 'Edit routes' page in the AWS Management Console for a VPC in the Mumbai region. The page displays a table of routes with columns for Destination, Target, Status, Propagated, and Route Origin. The first route is for destination 11.0.0.0/16, targeting 'local', with a status of 'Active'. Below this, there are three more routes: one for 0.0.0.0/0 targeting 'Internet Gateway', one for 10.0.0.0/16 targeting 'Peering Connection', and one for pcx-01646ef6965f09fef targeting 'Peering Connection'. Each of these three routes has a status of '-' and is not propagated. At the bottom right, there are buttons for 'Cancel', 'Preview', and 'Save changes'.

| Destination           | Target             | Status | Propagated | Route Origin     |
|-----------------------|--------------------|--------|------------|------------------|
| 11.0.0.0/16           | local              | Active | No         | CreateRouteTable |
| 0.0.0.0/0             | Internet Gateway   | -      | No         | CreateRoute      |
| 10.0.0.0/16           | Peering Connection | -      | No         | CreateRoute      |
| pcx-01646ef6965f09fef | Peering Connection | -      | No         | CreateRoute      |

Then this is the Peering of the two VPC's From Different Regions.

This is the requesting form from the N.virginia Region.

The screenshot shows the 'Create peering connection' page in the AWS Management Console for the N. Virginia region. The page is titled 'Select a local VPC to peer with'. It shows the 'VPC ID (Requester)' as vpc-084f2a0a522890979 (myvpc). Below this, it shows the 'VPC CIDRs for vpc-084f2a0a522890979 (myvpc)' with a table showing CIDR 10.0.0.0/16 with a status of 'Associated'. The page also has sections for 'Select another VPC to peer with', 'Account' (My account), 'Region' (Another Region), and 'VPC ID (Acceptor)' (vpc-08ce3f1ce9b299cdc).

| CIDR        | Status     | Status reason |
|-------------|------------|---------------|
| 10.0.0.0/16 | Associated | -             |

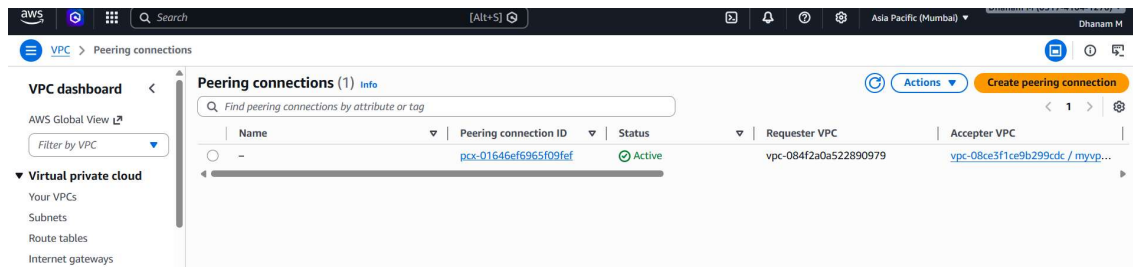
This is the Acceptor form in the Mumbai Region.

The screenshot shows the 'Accept VPC peering connection request' dialog box. It asks 'Are you sure you want to accept this VPC peering connection request? (pcx-01646ef6965f09fef)'. The dialog displays details for the Requester VPC (vpc-084f2a0a522890979), the Acceptor VPC (vpc-08ce3f1ce9b299cdc / myvpc1), the Requester CIDRs (10.0.0.0/16), the Acceptor CIDRs (empty), the Requester owner ID (051741041276), and the Acceptor owner ID (051741041276). At the bottom right, there are buttons for 'Cancel' and 'Accept request'.

| Requester VPC         | Acceptor VPC                   | Requester CIDRs |
|-----------------------|--------------------------------|-----------------|
| vpc-084f2a0a522890979 | vpc-08ce3f1ce9b299cdc / myvpc1 | 10.0.0.0/16     |

| Acceptor CIDRs | Requester Region        | Acceptor Region     |
|----------------|-------------------------|---------------------|
| -              | N. Virginia (us-east-1) | Mumbai (ap-south-1) |

| Requester owner ID         | Acceptor owner ID          |
|----------------------------|----------------------------|
| 051741041276(This account) | 051741041276(This account) |



We are using ping command to check whether the internet is connected with the two VPC.

1<sup>st</sup> Instance.



2<sup>nd</sup> Instance.

